

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

53001230 - Complements Of Thermodynamics

DEGREE PROGRAMME

05AZ - Master Universitario En Ingenieria Industrial

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	53001230 - Complements Of Thermodynamics
No of credits	3 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	05AZ - Master Universitario en Ingeniería Industrial
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Ignacio Lopez Paniagua		ignacio.lopez@upm.es	- -
Carlos Rafael Arnaiz Del Pozo (Subject coordinator)		cr.arnaiz@upm.es	- -

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

2.2. Research assistants

Name and surname	Email	Faculty member in charge
Roncal Casano, Juan Jose	juanjose.roncal@upm.es	Lopez Paniagua, Ignacio
Arnaiz Del Pozo, Carlos Rafael	cr.arnaiz@upm.es	Lopez Paniagua, Ignacio

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Química general
- Termodinámica básica

4. Skills and learning outcomes *

4.1. Skills to be learned

(a) - APLICA. Habilidad para aplicar conocimientos científicos, matemáticos y tecnológicos en sistemas relacionados con la práctica de la ingeniería.

(c) - DISEÑA. Habilidad para diseñar un sistema, componente o proceso que alcance los requisitos deseados teniendo en cuenta restricciones realistas tales como las económicas, medioambientales, sociales, políticas, éticas, de salud y seguridad, de fabricación y de sostenibilidad.

(e) - RESUELVE. Habilidad para identificar, formular y resolver problemas de ingeniería.

(h) - ENTIENDE LOS IMPACTOS. Educación amplia necesaria para entender el impacto de las soluciones ingenieriles en un contexto social global.

4.2. Learning outcomes

RA4 - Determinar efectos calóricos en sistemas reactivos.

RA127 - El alumno es capaz de organizar y dirigir su aprendizaje de forma autónoma para ampliar sus conocimientos en una materia.

RA118 - Ejecutar el procedimiento previsto. Valoración y validación del resultado obtenido.

RA116 - Identificar, analizar, e interpretar los datos del problema planteado por el profesor.

RA117 - Plantear un procedimiento/método de resolución.

RA119 - Valoración y validación del resultado obtenido.

RA125 - Utiliza correctamente técnicas de comunicación oral.

RA124 - Gestiona el tiempo de la presentación

RA2 - Determinar propiedades termodinámicas de mezclas.

RA1 - Aplicar los Principios de la Termodinámica Clásica en sistemas abiertos.

RA121 - Organiza la información.

RA3 - Resolver problemas de equilibrio físico en sistemas polifásicos.

RA5 - Resolver problemas de equilibrio químico en sistemas monorreactivos y monofásicos.

RA6 - Relacionar las propiedades macroscópicas con las microscópicas.

RA7 - Determinar las propiedades termodinámicas significativas y la eficiencia en ciclos directos e inversos.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Thermodynamics has three objectives for the training of an engineer:

Carry out energy balances to open systems undergoing physical processes (conservation of energy, First Law)

Analyse the quality of the energy flows in processes, either in terms of entropy or exergy (degradation of energy, Second Law)

Calculate thermophysical properties in systems with changes in composition: pure substances, mixtures, multiphase systems (vapour-liquid), reactive systems, etc.

This subject develops the concepts and tools required for calculating energy and exergy (entropy) balances in processes involving open systems with pure substances, mixtures, chemical reactions, etc that reach thermodynamic equilibrium.

The structure of the subject is developed based on three underlying pillars: Principles, Properties and Processes, which are applied to/calculated for open systems, i.e., those that exchange energy and mass with their surrounding.

5.2. Syllabus

1. Thermodynamic Principles to open systems
 - 1.1. General balances. Flow exergy
 - 1.2. Applications to steady state processes.
2. Properties of multicomponent systems
 - 2.1. Homogeneous systems
 - 2.2. Ideal mixture models.
 - 2.3. Real mixture models
3. Systems with composition change
 - 3.1. Equilibrium in multicomponent heterogeneous and reactive systems
 - 3.2. Multicomponent heterogeneous systems (VLE)
 - 3.3. Reactive systems
4. Thermodynamic Cycles
 - 4.1. Rankine cycles
 - 4.1.1. Direct
 - 4.1.2. Inverse
 - 4.2. Brayton cycle
 - 4.3. Other cycles

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Open systems Duration: 02:00 Lecture			
2	Open systems Duration: 02:00 Lecture			
3	Open systems Duration: 02:00 Lecture			Assignment 1 Individual work Progressive assessment Not Presential Duration: 00:00
4	Properties of Multicomponent Systems Duration: 02:00 Lecture			
5	Properties of Multicomponent Systems Duration: 02:00 Lecture			
6	Properties of Multicomponent Systems Duration: 02:00 Lecture			Assignment 2 Individual work Progressive assessment Not Presential Duration: 00:00
7	Systems with change in composition Duration: 02:00 Lecture			
8	Systems with change in composition Duration: 02:00 Lecture			
9	Systems with change in composition Duration: 02:00 Lecture			
10	Systems with change in composition Duration: 02:00 Lecture			Assignment 3 Individual work Progressive assessment Not Presential Duration: 00:00
11	Thermodynamic cycles Duration: 02:00 Lecture			
12	Thermodynamic cycles Duration: 02:00 Lecture			

13	Thermodynamic cycles Duration: 02:00 Lecture			
14	Thermodynamic cycles Duration: 02:00 Lecture			Assignment 4 Individual work Progressive assessment Not Presential Duration: 00:00
15				
16				
17				Examen final Written test Progressive assessment and Global Examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Assignment 1	Individual work	No Presential	00:00	0%	4 / 10	(a) (c) (e) (h)
6	Assignment 2	Individual work	No Presential	00:00	0%	4 / 10	(a) (e)
10	Assignment 3	Individual work	No Presential	00:00	0%	4 / 10	(a)
14	Assignment 4	Individual work	No Presential	00:00	0%	4 / 10	
17	Examen final	Written test	Face-to-face	02:00	100%	5 / 10	(a) (c) (e) (h)

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Examen final	Written test	Face-to-face	02:00	100%	5 / 10	(a) (c) (e) (h)

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen final escrito extraordinario	Written test	Face-to-face	02:00	100%	5 / 10	(e) (h) (a) (c)

7.2. Assessment criteria

The heterogeneity of the student's background may require adaptation of the syllabus and evaluation methods. Progressive evaluation (PE) will be carried during lectures, individually.

Their form, content and timing will be adapted to how the subject develops.

Progressive evaluation (PE) activities will add up to two (2) points to the final exam

A minimum grade of 4 must be attained in the Final Exam to add the averaged Progressive evaluation (PE) results to the final grade.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Termodinámica	Bibliography	Libro de texto
Tablas y gráficos de Termodinámica	Bibliography	Datos y tablas necesarios para la asignatura
Moodle	Web resource	Materials distributed on the moodle platform